

ASCII MASTER FLOW – PANEL 1/12: Glacier budget

ACCUMULATION -> snowfall, avalanche and refreezing

ELA -> annual gain equals annual loss

ABLATION -> melt, sublimation and calving where relevant

NET BALANCE -> geometry responds with a lag

MUST REMEMBER: Glaciation proceeds through accumulation-ablation balance, ice deformation/basal motion, erosion, transport and deposition; cirque-arête-horn, U-shaped valley, moraine and outwash logic must connect to Himalayan mass balance, proglacial lakes and GLOF cascades.

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ASCII MASTER FLOW – PANEL 2/12: Retreat paradox

ICE PARTICLES -> continue moving downslope

TERMINUS -> receives forward ice flux

ABLATION > DELIVERY -> snout shifts up-valley

METRIC RULE -> flow direction differs from margin change

ASCII MASTER FLOW – PANEL 3/12: Erosion suite

PLUCKING -> jointed blocks removed

ABRASION -> polish, striae and rock flour

CIRQUE / ARETE / HORN -> headwall system

U-TROUGH / HANGING VALLEY -> valley-scale overdeepening

ASCII MASTER FLOW – PANEL 4/12: Fjord-ria firewall

FJORD -> glacial trough later flooded by sea

RIA -> river valley later drowned by sea

FJORD CUES -> deep water, steep walls, hanging valleys

RULE -> inundation preserves different pre-existing valleys

ASCII MASTER FLOW – PANEL 5/12: Depositional sorting

TILL / MORaine -> direct ice, generally unsorted

ESKER -> meltwater ridge in or under ice

OUTWASH -> braided meltwater, relatively sorted

DRUMLIN -> streamlined drift/till landform

ASCII MASTER FLOW – PANEL 6/12: Ice-mass family

GROUNDING -> glacier, ice cap, ice sheet

FLOATING -> shelf, iceberg and sea ice

SEA LEVEL -> grounded loss adds ocean mass

BUTTRESSING -> shelf loss can accelerate inland discharge

ASCII MASTER FLOW – PANEL 7/12: Himalayan map hooks

GANGOTRI -> Bhagirathi

ZEMU -> Teesta

BARA SHIGRI -> Chenab system

SIACHEN -> Nubra-Shyok-Indus system

ASCII MASTER FLOW – PANEL 8/12: Water-timing sequence

EARLY LOSS -> local melt contribution may rise

STORED ICE DECLINES -> buffering weakens

LEAN SEASON -> reliability risk grows locally

QUALIFIER -> monsoon, snow and groundwater vary by basin

ASCII MASTER FLOW – PANEL 9/12: Glacial-lake test

BASIN / DAM -> how water is impounded

SLOPE + ICE -> possible displacement trigger

DAM CONDITION -> overtopping, piping or erosion

DOWNSTREAM -> exposure converts hazard into risk

ASCII MASTER FLOW – PANEL 10/12: GLOF cascade

TRIGGER -> slope failure, inflow or instability

WAVE / OVERTOPPING -> dam erosion begins

BREACH -> sudden release + sediment

VALLEY -> constrictions and assets shape consequences

CLOSE DISTINCTION: Snowline, equilibrium-line altitude, firn line and glacier terminus are distinct; till is unsorted ice-laid sediment while outwash is meltwater-sorted, and a fjord is a drowned glacial trough rather than any steep inlet.

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ASCII MASTER FLOW – PANEL 11/12: Risk-reduction stack

SATELLITE INVENTORY -> changing lakes and slopes

FIELD ASSESSMENT -> dam material and thresholds

WARNING + DRILLS -> last-mile action

LAND USE + DESIGN -> reduce exposure and fragility

EVIDENCE LIMIT: Retreat varies by glacier and period; lake growth alone does not prove imminent outburst, while dam material, freeboard, slope instability, ice/rock avalanches and downstream exposure condition risk.

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ASCII MASTER FLOW – PANEL 12/12: Cryosphere answer spine

DEFINE -> glacier, lake, GLOF and risk

MAP -> glacier-basin relation

EXPLAIN -> budget, trigger and breach

QUALIFY -> heterogeneity, attribution and data date